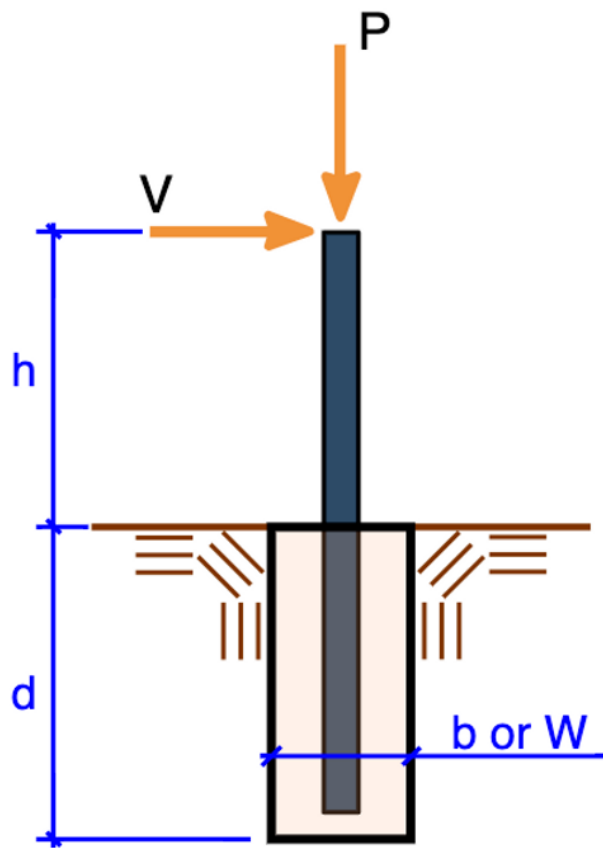




Pole Embedment

Example 2 - Flagpole Foundation

User Example

proj. 0002



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0.2 | Project description

Design of embedded pole foundation for the flagpole design example in Appendix A of NAAMM/FP 1001-07, “Guide Specifications for Design of Metal Flagpoles” Embedded pole foundation design is per 2024 IBC Eq 6-1 and Table 18-I-A. Soil properities are per Table 1806.2 in the 2024 IBC.

0.2 - 2 | Design input

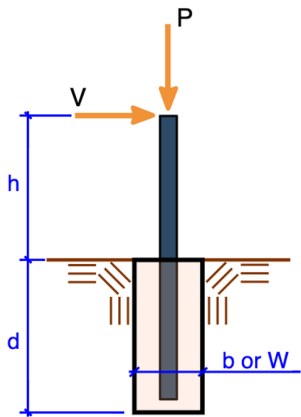


Fig. 1 - Calculation Diagram

Table 1: Design input

variable	value	[value]	description
Mbase	24.84 ft-kip	33.67 mkN	moment at base of flagpole
P	24.84 kips	110.47 kN	horizontal load
b	24.00 inch	60.96 cm	width of concrete drilled pier
PFP	200.00 pcf	31.42 kN_m3	allowable lateral bearing pressure - sandy gravel
h	1.50 ft	0.46 m	height
d	10.00 ft	3.05 m	initial guess for embedment depth

Table 2: Iterative functions (rvsrc/scripts/pole_embed.py)

Function	Docstring
Depth_nonconstrained(d, P, h, dia, PFP, tol)	Calculate required pole embedment using Eq. 18-1
Depth_constrained(d, P, h, dia, PFP)	Calculate required pole embedment using Eq. 18-2

0.2 - 3 | Design Results



Eq. 1: Required embed - nonconstrained

depth ₁ = Depth_nonconstrained(d, P, h, b, PFP, 2)				
depth ₁ = 15.52 ft [depth ₁] = 4.73 m Required embed - nonconstrained				
P	h	b	PFP	d
24.84 kips	1.50 ft	24.00 inch	200.00 pcf	10.00 ft
horizontal load	height	width of concrete drilled pier	allowable lateral bearing pressure - sandy gravel	initial guess for embedment depth
-	-	-	-	-

Eq. 2: Required embed - constrained

depth ₂ = Depth_constrained(d, P, h, b, PFP)				
depth ₂ = 7.34 ft [depth ₂] = 2.24 m Required embed - constrained				
P	h	b	PFP	d
24.84 kips	1.50 ft	24.00 inch	200.00 pcf	10.00 ft
horizontal load	height	width of concrete drilled pier	allowable lateral bearing pressure - sandy gravel	initial guess for embedment depth
-	-	-	-	-

